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Credit Card Fraud Detection Using Machine Learning Models

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Credit Card Fraud Detection
Using Machine Learning Models
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ABSTRACT: Credit card fraud detection has become increasingly vital as the rapid growth of online transactions continues to result in financial losses amounting to billions of dollars annually. Accurately identifying fraudulent transactions remains a significant challenge due to their infrequent and unpredictable nature. Fraudulent transactions typically represent less than 1% of all transactions, creating a highly imbalanced dataset. This imbalance hinders the performance of machine learning (ML) models, as they tend to favour the majority class, resulting in lower detection rates for fraud cases.

This project aims to evaluate the performance of three popular ML models: Logistic Regression, Random Forest (RF), and Extreme Gradient Boosting (XGBoost). The primary objective is to explore the impact of two key data balancing techniques—oversampling and undersampling—on the models' predictive accuracy and overall effectiveness. Oversampling involves generating synthetic samples for the minority class using methods such as the Synthetic Minority Over-sampling Technique (SMOTE). In contrast, undersampling reduces the majority class size to create a balanced dataset.

The models are trained and tested using a publicly available credit card fraud dataset, with performance assessed using key evaluation metrics such as Precision, Recall, F1-Score, and ROC-AUC. The study also examines trade-offs related to fraud detection accuracy, computational efficiency, false positive rates, and the potential for overfitting caused by oversampling techniques.

The results indicate that XGBoost with Tomek Links (undersampling) delivers the best performance, achieving a Precision of 0.94, an F1-Score of 0.84, and a ROC-AUC of 0.98. This demonstrates its strong capability in detecting fraudulent transactions while keeping false positives to a minimum. Random Forest also performs well, especially when combined with SMOTE, striking a balance between Precision and Recall. Conversely, Logistic Regression struggles with SMOTE, achieving high Recall but low Precision, highlighting the challenges of using linear models on highly imbalanced datasets.

Keywords: Credit Card Fraud Detection, Machine Learning Models, Data Imbalance, Oversampling, Undersampling, SMOTE, Tomek Links, Logistic Regression, Random Forest, XGBoost, Fraud Detection Systems, Financial Security.

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1. Introduction

In today's digital economy, credit card fraud detection has emerged as a critical concern due to the global shift toward cashless transactions. The rapid adoption of online banking, e-commerce platforms, and digital payment services has significantly increased the risk of financial fraud. Billions of dollars are lost annually to fraud, making credit card fraud one of the most prevalent and damaging forms of financial crime. This threat impacts millions of consumers and businesses worldwide, posing serious financial and reputational risks.

Fraud detection primarily relies on two key approaches: transaction monitoring and machine learning (ML). Both methods play essential roles in identifying suspicious activities, though they operate using distinct mechanisms and complement each other in modern fraud detection systems.

Transaction Monitoring involves continuously analysing transactions in real time to detect anomalies or unusual patterns based on predefined rules and thresholds. This approach flags irregular activities such as geographic location discrepancies, unusual purchase amount, unfamiliar merchant categories. However, rule-based systems can struggle with evolving fraud tactics, generating false positives or missing novel fraud patterns (IBM, 2024).

Machine Learning (ML) algorithms, on the other hand, learn fraud patterns from historical and real-time data, enabling dynamic and adaptive fraud detection. Unlike static rule-based systems, ML models continuously improve by adapting to new fraud trends through retraining. Common ML models used in fraud detection include logistic regression, random forests, gradient boosting, and deep learning models such as neural networks. These models excel at recognizing complex, non-linear fraud patterns, offering greater predictive accuracy.

Despite significant advancements in fraud prevention technologies, accurately detecting fraudulent transactions remains a persistent challenge. Fraud patterns evolve rapidly due to the sophisticated tactics employed by cybercriminals. As a result, financial institutions must adopt a proactive and adaptive approach, continuously updating and enhancing their detection systems to stay ahead of emerging threats while minimizing false positives and ensuring customer trust.

Background and Project Context

The motivation behind this project stems from the growing challenges faced by financial institutions in combating credit card fraud. The global shift toward digital transactions, driven by technological advancements and a connected economy, has resulted in increased financial crimes, including fraudulent credit card transactions.

Traditional fraud detection methods, such as rule-based systems and manual reviews, often struggle to keep up with the evolving tactics used by fraudsters. These static approaches fail to adapt to the dynamic and complex nature of modern fraudulent activities, necessitating more intelligent and adaptive solutions.

A critical challenge in this context is the severe data imbalance inherent in credit card transaction datasets. Fraudulent transactions typically constitute less than 1% of all transactions, leading to highly skewed datasets. Machine learning (ML) models trained on such imbalanced data are prone to favoring the majority class (legitimate transactions), resulting in high false-negative rates, where fraudulent transactions go undetected. Addressing this imbalance is essential to build robust and accurate fraud detection models.

Aims and Objectives

The primary aim of this project is to improve the detection of fraudulent credit card transactions by addressing data imbalance issues using advanced data balancing and machine learning techniques. To achieve this, the following specific objectives have been defined:

1. **Evaluate Model Performance:** Assess the performance of various machine learning models, including Logistic Regression, Random Forest, and XGBoost, in detecting fraudulent credit card transactions.
2. **Investigate Data Balancing Techniques:** Explore the impact of data balancing methods such as SMOTE (Synthetic Minority Oversampling Technique) and Tomek Links (undersampling) on model accuracy and fraud detection rates.
3. **Compare Model Performance Metrics:** Measure and compare models using key performance indicators such as Precision, Recall, F1-score, Accuracy, and ROC-AUC to identify the most effective approach.
4. **Analyze Trade-offs:** Examine trade-offs between detection accuracy, false-positive rates, model interpretability, and computational efficiency across various machine learning models and data-balancing techniques.

Methods Overview

To fulfill these objectives, a systematic research methodology was employed. The process began with data preprocessing, including data exploration, handling missing values, and data normalization. Data balancing techniques, specifically SMOTE and Tomek Links, were applied separately to create balanced datasets for training and evaluation.

The selected machine learning models—Logistic Regression, Random Forest, and XGBoost—were implemented and trained on pre-processed and balanced datasets. The evaluation phase involved assessing model performance using critical metrics such as Precision, Recall, F1-score, and ROC-AUC to measure the models' ability to distinguish between legitimate and fraudulent transactions. Additionally, computational efficiency and scalability were analyzed to ensure practical applicability in real-world fraud detection systems.

Report Structure Overview

The report is organized into five comprehensive chapters, each covering specific aspects of the research process:

Chapter 2: Literature Review: This chapter provides an extensive review of existing research on credit card fraud detection, emphasizing machine learning approaches and the challenges associated with data imbalance. It explores key concepts such as data balancing techniques (SMOTE and Tomek Links), commonly used classification algorithms, and evaluation metrics relevant to fraud detection.

Chapter 3: Data Exploration: This chapter focuses on exploring the dataset used for the study. It includes descriptive analysis, visualization of key features, and an examination of class distributions. Important steps such as handling missing values, detecting outliers, and performing feature selection are also discussed to provide a deeper understanding of the dataset's structure and potential challenges.

Chapter 4: Methodology: This chapter outlines the research methodology adopted for the project. It covers dataset selection, data preprocessing procedures, the implementation of data balancing techniques such as SMOTE and Tomek Links, and machine learning model configurations. The chapter also details the experimental framework, including hyperparameter tuning, model training, and evaluation criteria.

Chapter 5: Evaluation and Analysis: This chapter presents the experimental results and performance analysis of the models. It compares model performance across different data balancing techniques using key metrics such as Precision, Recall, F1-score, and ROC-AUC. Model-sampling combinations are assessed through statistical analysis and graphical representations, highlighting the trade-offs between detection accuracy and false positive rates.

Chapter 6: Conclusion and Future Work The final chapter summarizes the key findings of the project, emphasizing the best-performing models and discussing the broader implications of the results. It addresses project limitations, suggests possible enhancements, and offers recommendations for future research to further advance credit card fraud detection systems using cutting-edge machine learning approaches.

By systematically exploring data balancing techniques and evaluating machine learning models, this project aims to provide actionable insights for improving fraud detection frameworks. The findings offer practical guidance for financial institutions seeking to reduce financial losses and enhance transaction security in the digital financial ecosystem.

2. Literature Review

Introduction

The growing reliance on machine learning (ML) for critical applications such as fraud detection, medical diagnosis, and customer behaviour analysis has underscored the need for effective data preprocessing strategies. A particularly pressing challenge in ML is handling class-imbalanced datasets, where one class significantly outnumbers the other. This imbalance can skew a model's predictions toward the majority class, limiting its ability to identify minority class instances accurately. In credit card fraud detection, where fraudulent transactions constitute only a small fraction of all transactions, class imbalance can severely compromise the model's effectiveness, leading to increased financial losses and reduced system reliability.

Resampling techniques such as oversampling and undersampling have emerged as essential strategies for addressing class imbalance. Oversampling methods like the Synthetic Minority Oversampling Technique (SMOTE) generate synthetic samples for the minority class, while undersampling methods like Tomek Links remove borderline samples from the majority class. These approaches help restore balance in datasets, enhancing the model's ability to detect minority-class instances without sacrificing data integrity. Due to their effectiveness, these techniques have become widely adopted in fraud detection frameworks.

This literature review synthesizes recent research on the application of resampling techniques in credit card fraud detection. It explores key machine learning models, evaluation metrics, and data-balancing strategies, offering insights into best practices for optimizing fraud detection systems.

Class Imbalance and Its Impact

Class imbalance arises when the dataset contains significantly more samples from one class than the other. This imbalance biases the ML model toward the majority class, reducing its ability to predict the minority class accurately. Common applications affected by this issue include fraud detection, medical diagnostics, and customer behaviour analysis. The reviewed literature highlights that resampling methods are critical in mitigating this challenge by balancing datasets before model training (Mohammed, 2020).

Oversampling Methods

Oversampling addresses class imbalance by increasing the representation of the minority class through various techniques:

- **Random Oversampling:** This method duplicates existing minority samples to balance the dataset. However, it can cause overfitting by making the model memorize repeated samples.
- **SMOTE (Synthetic Minority Oversampling Technique):** A popular approach that generates synthetic samples by interpolating between nearest minority-class samples, enhancing the decision boundary.

- Borderline-SMOTE: This technique targets samples near the decision boundary, creating synthetic samples that reduce classification errors in complex datasets (Mohammed, 2020).
- Adaptive Synthetic Sampling (ADASYN): A dynamic variation of SMOTE that generates samples based on the learning difficulty of instances, improving classification performance (Mohammed, 2020).

Although oversampling enhances minority-class detection, it may induce overfitting, particularly in models with limited generalization capabilities.

Undersampling Methods

Undersampling reduces the size of the majority class by removing redundant samples, balancing the dataset. Key techniques include:

- Random Undersampling: This method randomly removes samples from the majority class, potentially causing information loss.
- Tomek Links: This approach removes overlapping samples near decision boundaries, improving class separation and model clarity.
- Cluster Centroids: Majority-class samples are replaced with centroids representing clusters, maintaining the dataset's overall structure (Mohammed, 2020).
- Near Miss: This method selects majority samples closest to minority samples, balancing the dataset while preserving relevant decision boundaries (Mohammed, 2020).

Though computationally efficient, undersampling risks data loss, potentially limiting the model's performance on highly imbalanced datasets.

1. Traditional Machine Learning Algorithms

- Logistic Regression (LR): Logistic regression is a commonly used binary classification model valued for its simplicity, interpretability, and computational efficiency. However, its reliance on linear decision boundaries makes it less effective when dealing with highly imbalanced datasets or complex relationships in fraud detection scenarios.
- Decision Trees (DT): Decision trees are transparent models that divide data into smaller subsets based on specific feature conditions. While easy to interpret, they are prone to overfitting, especially when the dataset is large or noisy. Proper regularization techniques such as pruning or limiting tree depth are essential to improve performance.
- Support Vector Machines (SVMs): SVMs excel at handling high-dimensional feature spaces by finding the optimal decision boundary between classes. However, their performance is sensitive to parameter tuning, such as selecting the correct kernel type and regularization parameter, making them computationally intensive for large datasets (Bhakta, 2023).
- Naive Bayes (NB): Naive Bayes is a computationally efficient classification model based on probabilistic reasoning. However, it assumes that features are independent, which may not hold in real-world credit card fraud detection datasets. This assumption can reduce its effectiveness when feature interactions are critical for accurate detection (Bhakta, 2023).

2. Ensemble Learning Techniques

Ensemble learning techniques improve prediction accuracy by combining multiple models, leveraging their strengths, and reducing individual model limitations. Key ensemble methods commonly used in credit card fraud detection include:

- **Random Forest (RF):** Random Forest is an ensemble of decision trees that operates by constructing multiple decision trees during training and averaging their predictions. This approach reduces variance and mitigates overfitting, making it highly effective for complex fraud detection tasks. Its robustness and ability to handle imbalanced datasets make it a popular choice for fraud detection systems.
- **XGBoost and Gradient Boosting:** These boosting algorithms build models sequentially by correcting errors from previous iterations. XGBoost, known for its speed and efficiency, applies regularization techniques to avoid overfitting. Both methods enhance prediction accuracy and precision by focusing on difficult-to-classify instances.
- **Voting and Stacking:** Voting combines predictions from multiple models by averaging their outputs (for regression) or using majority voting (for classification). Stacking goes further by training a meta-model on the predictions of base models, improving overall system performance through diverse learning strategies.
- **AdaBoost:** AdaBoost adjusts the weights of misclassified instances, making the model more sensitive to previously incorrect predictions. It sequentially trains weak classifiers, emphasizing challenging samples, and aggregates their outputs for improved classification accuracy (Bhakta, 2023).

3. Deep Learning and Anomaly Detection

- **Neural Networks (NNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) Networks:** These models capture complex temporal patterns but require extensive datasets and computational resources (Bhakta, 2023).
- **Anomaly Detection Models:** Models like Isolation Forest, Local Outlier Factor (LOF), and Generalized Isolation Forest detect unusual patterns in transaction data, supporting real-time fraud detection (Bhakta, 2023).

Evaluation Metrics

Given the class-imbalanced nature of credit card fraud detection, relying solely on accuracy can be misleading, as models may achieve high accuracy by predicting only the majority class. Therefore, more comprehensive evaluation metrics are essential to assess model performance effectively:

- **Accuracy:** Reflects the proportion of correct predictions made by the model. Although frequently used, it may be misleading when dealing with imbalanced datasets where one class significantly outnumbers the other.
- **Precision:** Represents the fraction of correctly identified fraud cases out of all cases flagged as fraud. A high precision score indicates fewer false positives, reducing unnecessary investigation efforts.

- **Recall (Sensitivity):** Measures the percentage of actual fraud cases successfully detected by the model. A higher recall score means fewer fraudulent transactions go unnoticed, minimizing false negatives.
- **F1-Score:** Calculates the harmonic mean of precision and recall, providing a balanced evaluation. This metric is particularly useful when both false positives and false negatives carry significant consequences.
- **ROC-AUC (Receiver Operating Characteristic - Area Under the Curve):** Assesses the model's ability to distinguish between fraudulent and non-fraudulent cases. A higher AUC score indicates better classification performance, even in scenarios with imbalanced datasets.

These metrics provide a more holistic evaluation of model performance, ensuring fraud detection systems are both accurate and sensitive to fraudulent activity. Let me know if you'd like additional details!

Conclusion

Resampling techniques play a crucial role in mitigating class imbalance in machine learning-based fraud detection. This research focuses on comparing the performance of key models such as Random Forest, XGBoost, and Logistic Regression when combined with two distinct resampling methods: Tomek Links and SMOTE. The comparative evaluation highlights how different resampling strategies influence model accuracy, recall, and overall detection performance in credit card fraud detection.

The findings emphasize that while SMOTE enhances the minority class representation through synthetic sampling, Tomek Links improves class separation by removing borderline instances. Understanding how these techniques affect various machine learning models helps identify optimal configurations for fraud detection frameworks.

Future research should explore advanced hybrid sampling methods, adaptive model tuning, and explainable AI techniques to enhance interpretability and real-time fraud detection capabilities, ensuring continuous improvement in combating financial fraud.

3. Legal, Social, Ethical, and Professional Issues

Fraud detection systems in financial services, e-commerce, and banking industries must address a range of legal, social, ethical, and professional concerns. These systems use sensitive personal and financial data, making compliance with data protection laws, ethical data use, and responsible algorithm deployment critical. This section highlights key considerations related to fraud detection models and data-driven decision-making processes.

1. Legal Issues

Fraud detection systems operate under strict legal frameworks to ensure data privacy, consumer protection, and regulatory compliance. Key legal concerns include:

- **Data Protection Laws:** Compliance with global data privacy regulations such as the General Data Protection Regulation (GDPR) in the EU and the California Consumer Privacy Act (CCPA) in the US is essential. These laws regulate how personal data is collected, processed, stored, and shared (DataGuidance, 2022).
- **Consumer Rights:** Users have the legal right to be informed about how their data is used. Fraud detection systems must provide transparency and enable users to contest incorrect classifications, such as false fraud alerts leading to financial service denials (Okta, 2021).
- **Anti-Money Laundering (AML) and Know Your Customer (KYC):** Businesses must comply with financial crime laws, including AML and KYC regulations, to prevent fraud, terrorism financing, and money laundering activities. Non-compliance can result in severe financial penalties and legal repercussions (AUSTRAC, 2024).

2. Social Issues

Fraud detection models have significant social implications due to their impact on individuals, businesses, and society as a whole:

- **Bias and Discrimination:** Algorithmic bias can result from imbalanced datasets or biased features, leading to unfair outcomes. For example, if a fraud detection model unfairly flags specific demographic groups as fraudulent due to historical biases, it may cause social harm.
- **Social Trust and Reputation:** Businesses that use fraud detection systems risk damaging their reputation if false positives (incorrect fraud alerts) or false negatives (missed fraud cases) occur frequently. This can erode customer trust and impact long-term customer loyalty.
- **Economic Impact:** Widespread financial fraud affects the economy by causing financial losses, reducing consumer confidence, and driving up the cost of doing business. Effective fraud detection mitigates these issues, benefiting society.

3. Ethical Issues

Fraud detection raises ethical concerns around the fair and transparent use of algorithms and personal data:

- **Transparency and Explainability:** Machine learning models like XGBoost and Random Forest are often seen as “black boxes” (Ambiata, 2021) due to their complexity. Ethical concerns arise when businesses cannot explain how fraud detection decisions are made. This lack of transparency can reduce public trust and conflict with ethical AI principles.
- **Data Consent and Ownership:** Collecting and processing user data for fraud detection must be based on informed consent. Users should know how their data is used and have the right to opt out when possible.
- **Data Misuse and Surveillance Risks:** Companies must avoid data misuse, such as surveillance practices or sharing customer data without proper authorization. Ethical breaches can result in public backlash and long-term reputational damage.
- **Accountability and Fairness:** Businesses deploying fraud detection systems must take responsibility for false positives and negatives. Misclassifications can deny customers access to essential financial services, raising concerns about fairness and accountability.

4. Professional Issues

Developers, data scientists, and financial analysts working in fraud detection must adhere to professional standards to ensure system reliability and accountability:

- **Data Security and Integrity:** Professionals must follow best practices for data security, including encryption, secure data storage, and access control policies. This prevents unauthorized access to sensitive personal and financial information.
- **Ethical Development Practices:** Fair model design, unbiased data selection, and ethical AI practices should be embedded in the development process. This includes conducting bias audits, ethical impact assessments, and using interpretable models where possible.
- **Compliance Audits and Reporting:** Regular compliance audits ensure that fraud detection systems adhere to regulatory standards and operate ethically and professionally. Reports should be transparent and shared with internal and external stakeholders.

4. Data Exploration

The dataset contains credit card transactions made by European cardholders in September 2013. It includes transactions over a two-day period, comprising a total of 284,807 transactions, of which 492 are fraudulent, representing approximately 0.172% of all transactions. This highlights the significant class imbalance within the dataset, where fraudulent transactions (positive class) constitute a very small fraction compared to legitimate transactions (negative class).

The dataset consists exclusively of numerical features, which are the result of a Principal Component Analysis (PCA) transformation. To maintain confidentiality, the original features and additional background information have not been provided. The features are structured as follows:

- V1, V2, ..., V28: These are the principal components generated through PCA.
- Time: This feature indicates the number of seconds that have passed between each transaction and the first recorded transaction in the dataset. It was not subjected to PCA transformation.
- Amount: This feature indicates the transaction amount and can be used for tasks such as cost-sensitive learning.
- Class: This is the target variable, where a value of 1 indicates a fraudulent transaction, and a value of 0 indicates a legitimate transaction.

Overall, this dataset is particularly challenging due to its extreme imbalance and the transformed nature of most features, making it ideal for evaluating machine learning models and techniques designed to handle imbalanced classification problems.

Data Exploration

1. Data Exploration

1.1 Exploring the Dataset

The first step in understanding the dataset involves examining its structure, columns, and statistical properties. The dataset used for this project contains transaction records with anonymized features to protect sensitive information, it includes 28 numerical features labeled as V1 to V28, which are the result of Principal Component Analysis (PCA). In addition, there are two important undeformed features: Time and Amount. The target variable, Class, indicates whether a transaction is fraudulent (1) or legitimate (0).

To better understand the data, a statistical summary was generated using the `.describe()` function showing on the Table 1 below. The time feature represents the elapsed seconds since the first transactions and spans approximately 48 hours (172,792 seconds). The amount features show a large variation, with most transaction amounts being small, but a few reaching extreme values. The target variable confirms that the dataset is binary, containing two distinct classes.

Table 1: A statistical summary

Feature	Mean	STD	Min	25%	50%	75%	Max
Time	94813.86	47488.15	0	54201.50	84692.00	139320.50	172792.00
Amount	88.35	250.12	0	5.60	22.00	77.166	25691.16

1.2 Checking for Missing Data

An essential part of the exploration process is checking for missing or null values, as they can hinder the effectiveness of machine learning models. Using the `isnull().sum()` function, the dataset was examined for missing data. The results confirms that there are no missing data or null values in any of the features. This ensures that all features are complete and ready for further analysis and preprocessing

1.3 Key Features: Time and Amount

1.3.1 Feature Distributions and Boxplots

The Time and Amount features are important as they provide insights into transaction patterns and outliers.

The Time feature was initially recorded in seconds but was transformed into Hour of Day for better interpretability. A Kernel Density Estimation (KDE) plot was generated to visualize the distribution of fraudulent and non-fraudulent transactions over the 24-hour period. The results show that fraudulent transactions tend to cluster during specific hours, particularly between midnight and early morning, while legitimate transactions are more uniformly distributed throughout the day.

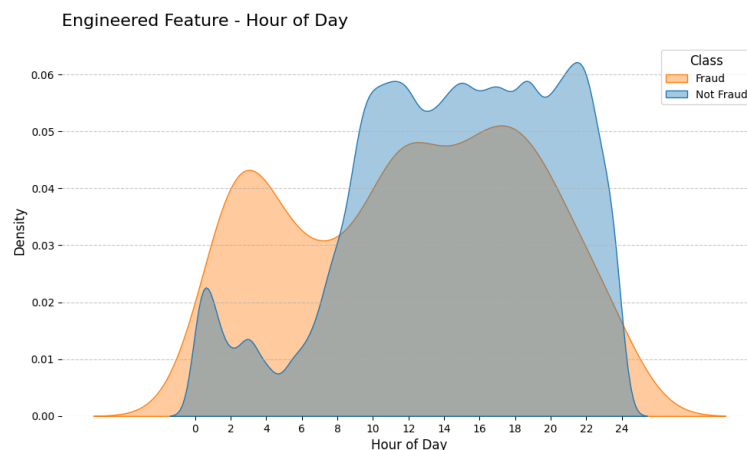


Figure 1: KDE plot showing the Hour of Day distribution for fraud and non-fraud classes.

The Amount feature reveals a right-skewed distribution. Most transactions involve relatively small amounts, with a median value of 22. However, several transactions exhibit extremely high values, reaching up to 25,691. These high-value transactions were further analysed using a boxplot, which highlights the presence of significant outliers. Outliers in the Amount feature

are critical for fraud detection, as fraudulent activities often involve unusually large or unusually small transaction amounts.

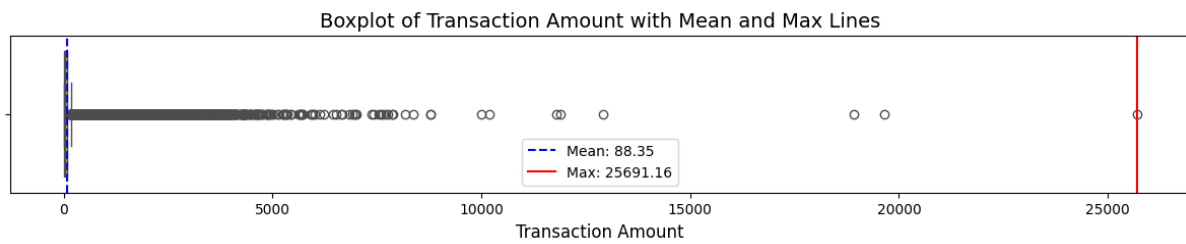


Figure 2 Boxplot of feature “Amount”

1.4 Class Distributions

The class distribution in the dataset reveals a substantial imbalance between fraudulent and non-fraudulent transactions. Fraudulent transactions, represented by Class 1, account for only 0.17% of the total data, while legitimate transactions (Class 0) dominate with 99.83%. This severe class imbalance poses a significant challenge for machine learning models, as they may be biased toward predicting the majority class while failing to detect fraud.

A bar chart was used to visualize the class distribution on Figure 3, further confirming the imbalance. Addressing this issue is a critical aspect of the project, and it will be managed through techniques such as SMOTE (Synthetic Minority Over-sampling Technique) and Tomek Links (undersampling) to balance the dataset and improve model performance.



Figure 3

1.5 Correlation Analysis

To identify relationships between features and evaluate their importance for model input, a correlation matrix was computed using Pearson's correlation coefficient. Pearson's coefficient measures the linear relationship between two variables, with values ranging from -1 to 1, where:

- A positive value signifies a direct correlation, meaning both variables increase together.
- A negative value denotes an inverse correlation, where one variable increase while the other decreases.

The dataset comprises 28 anonymized numerical features (V1 to V28) derived through Principal Component Analysis (PCA), as well as two untransformed features: Amount and Time. A heatmap was generated using Seaborn to visualize the pairwise correlations, providing an intuitive understanding of these relationships, as shown in Figure 4.

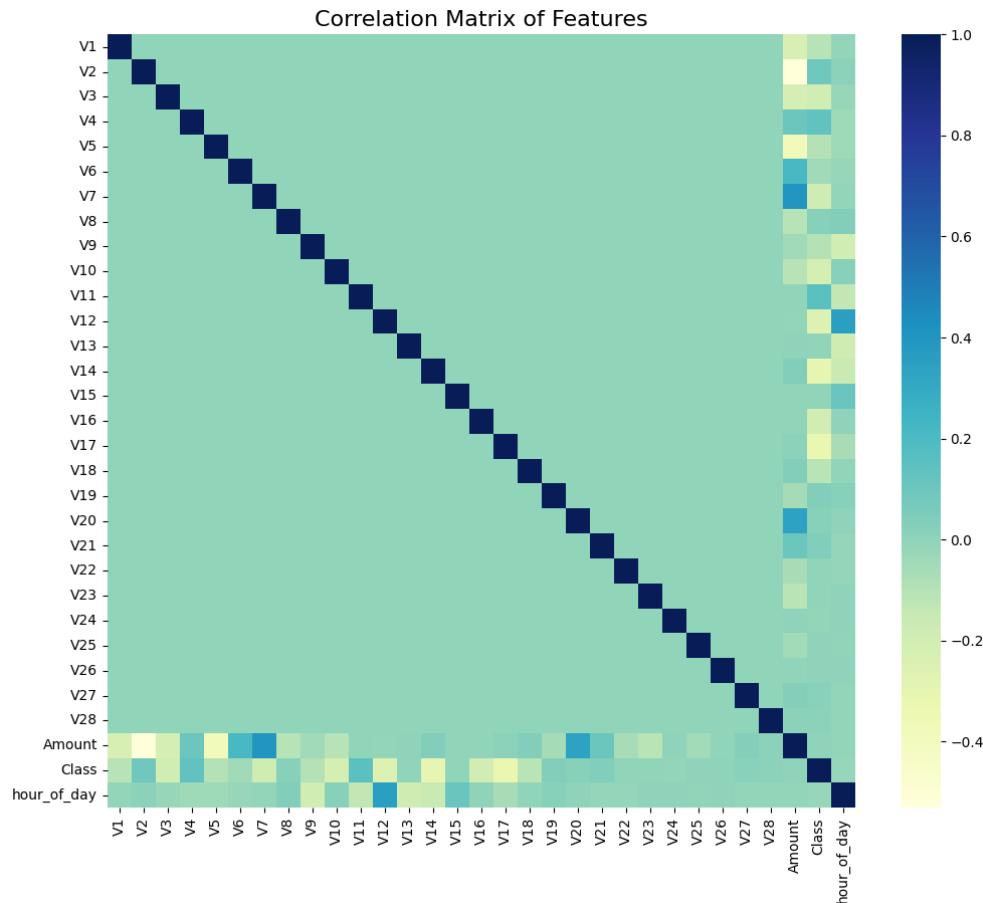


Figure 4 Correlation Heatmap

The analysis focused specifically on the correlation between the features and the target variable (Class), which indicates whether a transaction is fraudulent (1) or legitimate (0). Features with the strongest positive or negative correlations were identified for prioritization in the machine learning process.

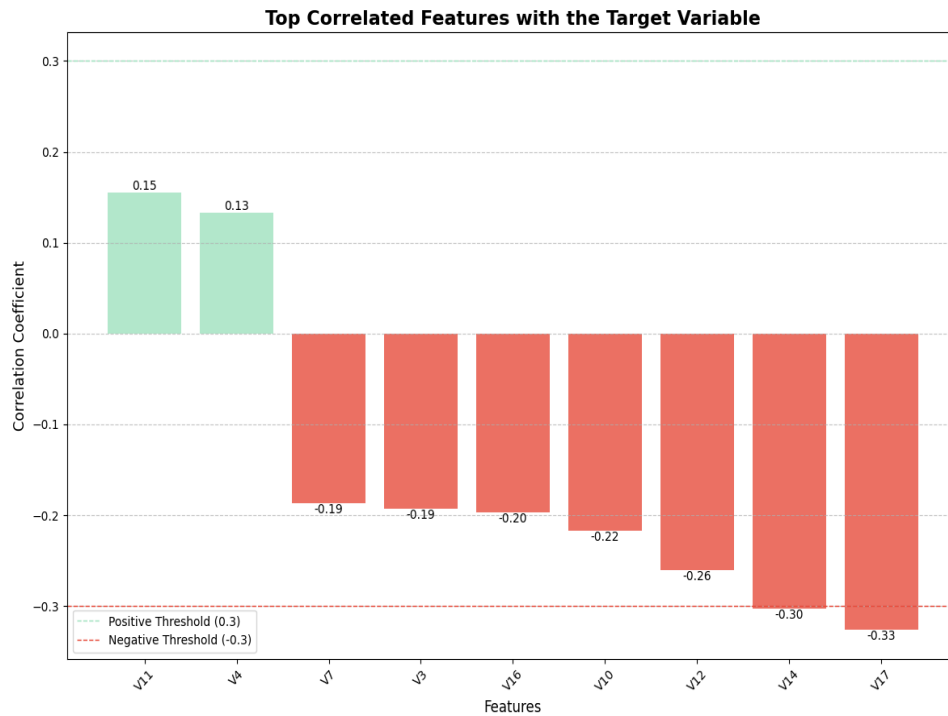


Figure 5 high correlation features

As highlighted in Figure 5, the features V11, V4, V17, V14, and V12 demonstrate relatively stronger correlations with the target variable when compared to other features.

Positive Correlations: V11 (0.15) and V4 (0.13) exhibit the strongest positive correlations with the target variable. This suggests that as the values of these features increase, the likelihood of a transaction being fraudulent may also increase, although the relationship remains weak.

Negative Correlations: V17 (-0.33), V14 (-0.30), and V12 (-0.26) show the strongest negative correlations with the target variable. This indicates that lower values of these features are more associated with fraudulent transactions.

Among these, V17 stands out as having the highest negative correlation, making it a particularly important feature for distinguishing fraudulent transactions.

Implications for model training

The features with stronger correlations, such as V17, V14, and V11, are expected to contribute significantly to the performance of machine learning models. While these correlations are relatively weak in absolute terms, they still offer valuable information for detecting fraud. Features with weaker correlations may have limited standalone predictive power but can still provide value when used in conjunction with other features in non-linear models such as Random Forest and XGBoost.

1.6 Data Splitting

To prepare the dataset for model training and evaluation, the data was split into training and testing sets following a structured process. This step ensured that the models were trained effectively while being evaluated on unseen data to gauge their performance and generalization capabilities.

Feature Selection

Initially, features with high correlations identified through correlation analysis were selected as input features for the models. The features chosen included V11, V4, V17, V14, and V12, which demonstrated relatively strong relationships with the target variable (Class). These features were prioritized due to their predictive significance, contributing valuable information for detecting fraudulent transactions. By focusing on these key features, the dataset's complexity was reduced while retaining the most informative predictors for fraud detection. This feature selection process helps improve model interpretability, reduce overfitting risks, and enhance computational efficiency during training.

Dataset Splitting Process

The dataset was then divided into training and testing subsets using an 80:20 split ratio:

- 80% of the data was assigned to the training set, enabling the models to learn fraud detection patterns through exposure to a large portion of the dataset.
- 20% of the data was reserved for the test set to evaluate the model's performance on unseen data, ensuring reliable performance assessment.

5. Methodology

2.1 Feature Scaling

Feature scaling is a crucial preprocessing step in machine learning, particularly when features have different ranges or units. Scaling ensures that all features contribute equally to the model, avoiding biases caused by large-scale features and improving the performance of distance-based models like logistic regression or tree-based models such as Random Forest.

2.1.1 Robust Scaler

The amount feature, which represents the transaction value, contains extreme outliers (as observed in the boxplot). These outliers can skew the scaling process, resulting in poor model performance. To address this, Robust Scaler was applied. The Robust Scaler standardizes data by utilizing the median and interquartile range (IQR), making it resistant to the influence of outliers. The scaling process follows this formula:

$$X_{scaled} = \frac{X - Q_2}{Q_3 - Q_1}$$

Equation 1 Robust Scaler

Where:

- Q_1 is the 25th percentile (first quartile),
- Q_2 is the median value,
- Q_3 is the 75th percentile (third quartile).

This approach ensures that the Amount feature is rescaled effectively. Without being influenced by extreme values

2.1.2 Standard Scaler

The Time feature represents the number of seconds elapsed since the first transaction. Since this feature does not exhibit extreme outliers, StandardScaler was chosen to normalize its range. StandardScaler transforms the data to have a mean of 0 and a standard deviation of 1 using the formula:

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

Equation 2 Standard Scaler

Where:

- X is the original value,
- μ is the mean of the feature,
- σ is the standard deviation of the feature.

This ensures the Time feature is standardized, allowing it to contribute appropriately to the machine learning models

2.2 Handling Imbalanced Data

The dataset used in this study exhibits a significant class imbalance, with fraudulent transactions constituting less than 1% of the total records. This imbalance poses a critical challenge in developing effective fraud detection models, as standard machine learning algorithms tend to favour the majority class, leading to biased predictions and a high rate of false negatives (missed fraud cases). Consequently, specialized techniques were applied to address this issue and improve model performance.

To mitigate the effects of class imbalance, two resampling methods were employed:

2.2.1 Oversampling using SMOTE

SMOTE (Synthetic Minority Over-Sampling Technique) is a data augmentation method aimed at addressing class imbalances by generating synthetic samples for the minority class. Unlike basic oversampling methods that duplicate existing samples, SMOTE creates new instances through interpolation, enhancing the diversity of the minority class representation.

The SMOTE process includes the following steps:

1. **Finding Nearest Neighbors:** Identify the k-nearest neighbors for each minority-class sample using the K-Nearest Neighbors (KNN) algorithm.
2. **Generating Synthetic Samples:** Create synthetic data points by interpolating between the original minority-class samples and their nearest neighbors using a randomly chosen interpolation factor.
3. **Updating the Dataset:** Add the synthetic samples to the original dataset, resulting in a more balanced class distribution.

By increasing the representation of the minority class, SMOTE reduces overfitting and enhances the machine learning model's ability to generalize, particularly when detecting rare events like fraudulent transactions.

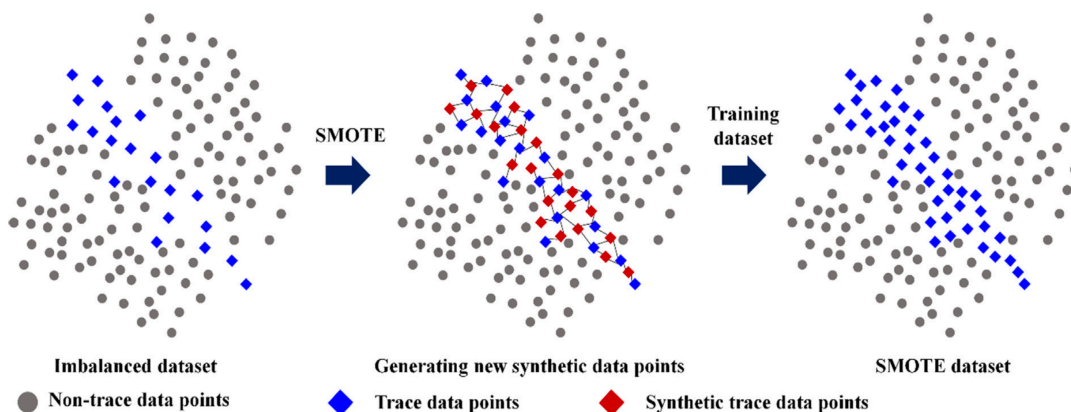


Figure 6: Visualization of the SMOTE Process: The SMOTE algorithm generates synthetic samples (red diamonds) between existing minority-class points (blue diamonds), balancing the dataset for training.

Below is the comparison of the dataset before and after applying SMOTE.

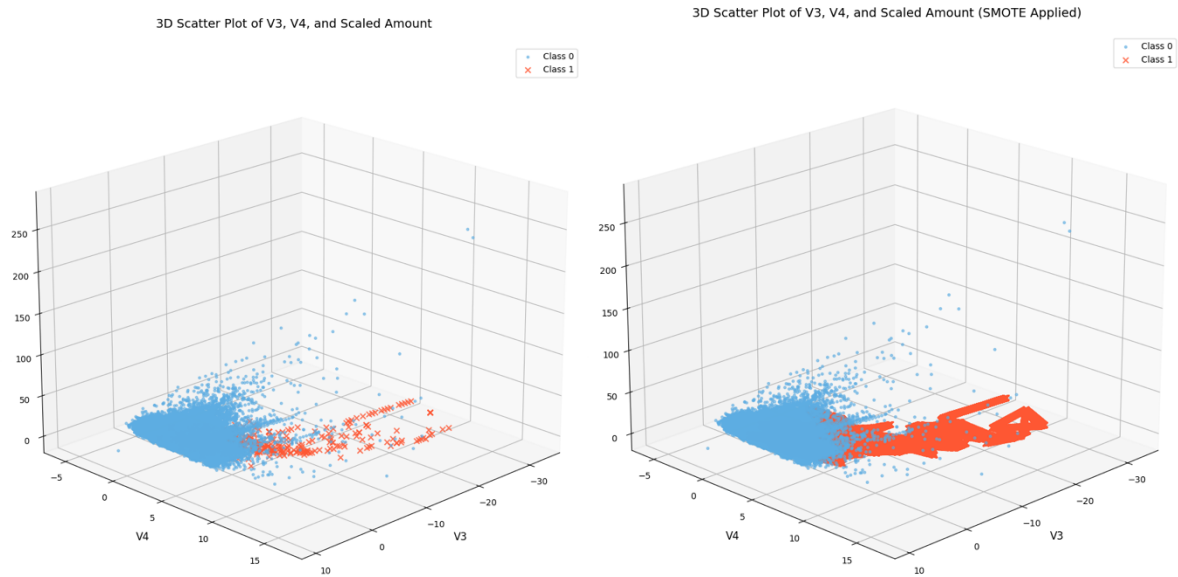


Figure 7 Comparison of training data before and after applying SMOTE

2.2.2 Undersampling using Tomek Links

Tomek Links is an undersampling technique that focuses on cleaning the majority class by removing borderline or overlapping samples. A Tomek Link exists between two samples X_i (minority class) and X_j (majority class) if:

1. X_j is the closest neighbour of X_i ,
2. X_i is also the closest neighbour of X_j .

By removing the majority-class sample X_j from Tomek Links, this technique improves the decision boundary between classes.

Advantages of Tomek Links:

- Reduces the size of the majority class while preserving informative samples.
- Improves the model's ability to distinguish between classes.
-

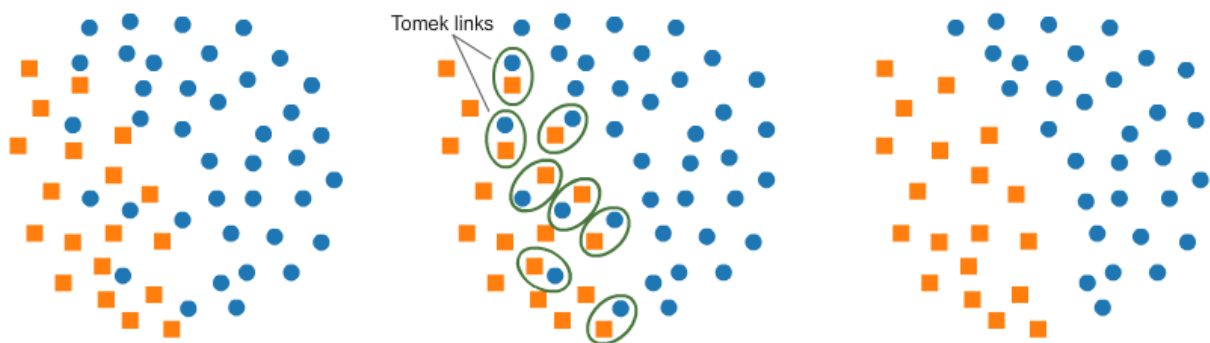


Figure 8 Tomek Links

2.3 Model Selection and Training

To identify fraudulent transactions effectively, three machine learning models were selected and trained: Logistic Regression, Random Forest, and XGBoost. Each model was implemented using the reprocessed dataset, including feature scaling and handling of class imbalance (e.g., SMOTE, Tomek Links, and undersampling).

The steps involved in training include:

1. Model initialization
2. Hyperparameter tuning to optimize performance.
3. Training the models on the balanced training dataset.

2.3.1 Logistic Regression

Logistic Regression (LR) is a fundamental statistical model widely used for binary classification tasks such as fraud detection. It estimates the probability of a binary outcome (fraud or non-fraud) based on a linear combination of input features. The simplicity, interpretability, and computational efficiency of Logistic Regression make it a common baseline model for classification problems, including those involving imbalanced datasets.

Working Mechanism

Logistic Regression works by modelling the relationship between independent variables (features) and the log-odds of the target variable (fraud or non-fraud). The model uses the logistic function (sigmoid function) to map predicted values to probabilities between 0 and 1:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

Equation 3 Logistic Regression

Based on the equation 3 above, p represents the predicted probability of fraud, X_n are the input features, β_n are the feature coefficients, and β_0 is the intercept. The model is trained by minimizing the log-loss (cross-entropy loss) function, which penalizes incorrect predictions more heavily when the predicted probability deviates significantly from the true label.

Advantages of Logistic Regression

Logistic Regression is highly interpretable, as the coefficients of the model indicate the impact of each feature on the likelihood of fraud. This makes it particularly useful for understanding the contribution of different factors in fraud detection. Additionally, its efficiency allows it to scale well to large datasets, and it is less prone to overfitting when paired with regularization techniques. Despite its simplicity, Logistic Regression is effective for imbalanced datasets when used with techniques like SMOTE and Tomek Links, as emphasized in the literature.

The literature establishes Logistic Regression as a baseline model for classification tasks due to its interpretability and computational efficiency. For fraud detection, it serves as a strong benchmark, particularly when paired with resampling methods like SMOTE and Tomek Links.

While more complex models like Random Forest and XGBoost often outperform Logistic Regression in detecting fraudulent transactions, LR provides valuable insights into feature importance and serves as a lightweight option for quick analysis.

2.3.2 Random Forest

Random Forest is a robust and flexible machine learning algorithm commonly used for both classification and regression tasks. As an ensemble learning technique, it constructs multiple decision trees and aggregates their predictions to enhance accuracy and model stability (Breiman, 2020).

Working Mechanism

Random Forest combines multiple decision trees through an ensemble approach to improve performance and robustness. The key steps in the Random Forest workflow are as follows:

1. **Bootstrap Sampling:** Multiple decision trees are trained on randomly sampled subsets of the training data.
2. **Feature Selection:** At each split, only a subset of features is considered, reducing correlation between trees.
3. **Tree Building:** Each tree is trained independently using the chosen data and features.
4. **Voting Mechanism:** For classification tasks, the final prediction is based on the majority vote of all trees.

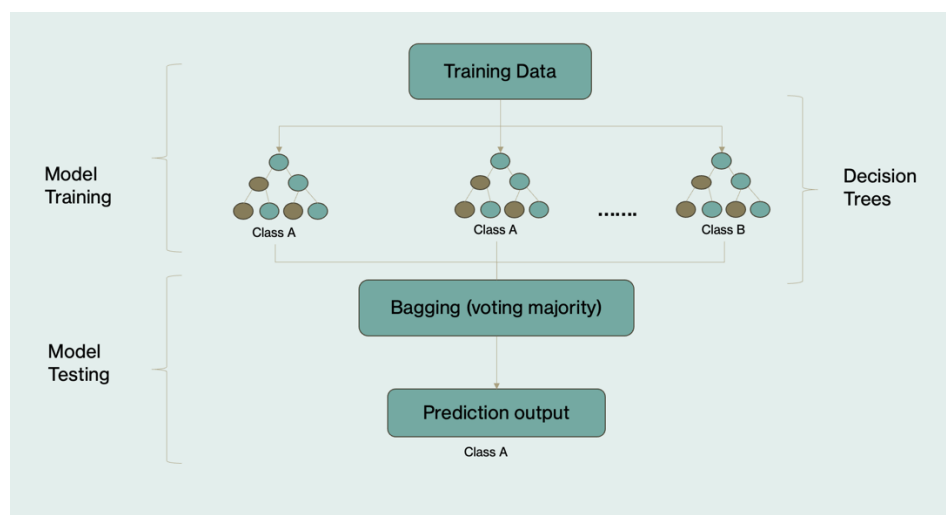


Figure 9 Random Forest Diagram.

2.3.3 XGBoost

XGBoost (Extreme Gradient Boosting) is a highly efficient and scalable machine learning algorithm designed for supervised learning tasks such as regression, classification, and ranking. Developed by Tianqi Chen in 2016, XGBoost has gained widespread adoption due to its superior performance, computational efficiency, and ability to handle complex, high-dimensional datasets.

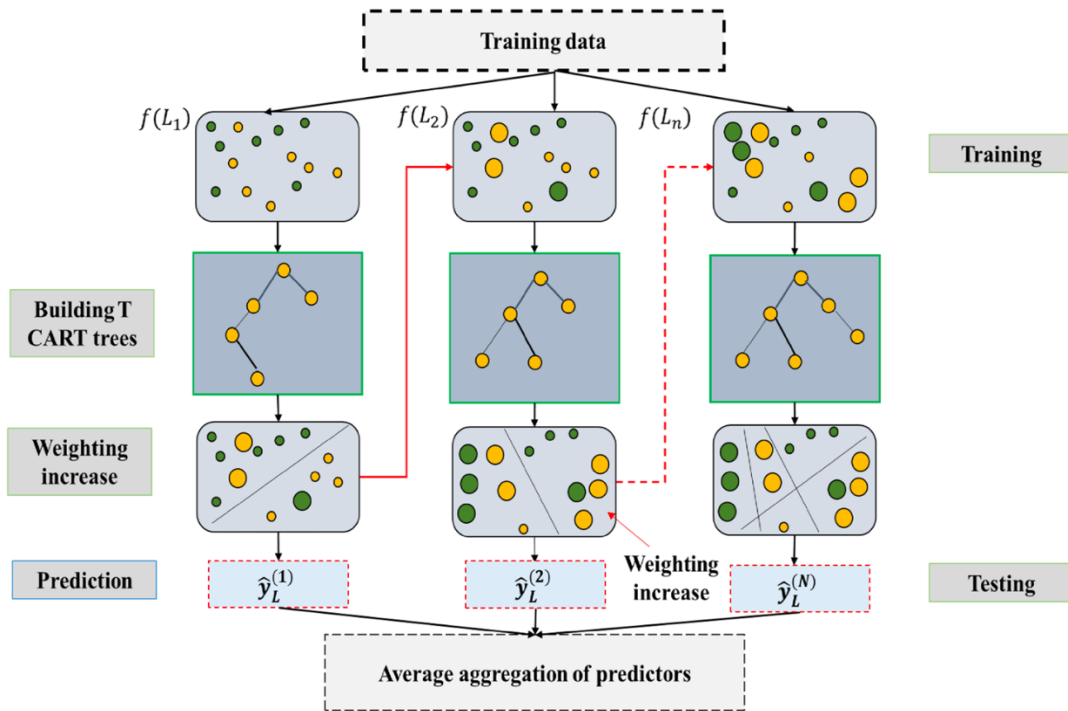


Figure 10 XGBoost Diagram

The figure 10 above illustrates the boosting process in ensemble learning, where multiple weak learners (CART trees) are trained iteratively to improve classification performance. The process begins with training data where all samples are initially weighted equally. After the first tree ($f(L_1)$) is trained, misclassified samples are assigned higher weights, ensuring the next tree ($f(L_2)$) focuses more on these difficult instances. This weighting adjustment continues through subsequent iterations ($f(L_3)$, ..., $f(L_n)$), with each tree improving on the errors of its predecessor. Once all trees are trained, predictions from each tree are aggregated using weighted averaging to make the final prediction. This iterative process enhances model accuracy by emphasizing hard-to-classify samples, making boosting techniques like AdaBoost or Gradient Boosting effective for complex tasks such as fraud detection.

Research consistently highlights XGBoost as a top-performing model for tasks such as fraud detection due to its scalability, robust regularization, and ability to manage class imbalance. Studies have shown that XGBoost often outperforms traditional models like Logistic Regression and Random Forest, especially when dealing with highly imbalanced datasets.

Model fitting and Hyperparameters for Fraud Detection

To optimize performance in fraud detection tasks, specific hyperparameters were tuned for Random Forest, Logistic Regression, and XGBoost. These hyperparameters were carefully selected based on their impact on model accuracy, recall, and computational efficiency, ensuring the models perform well in detecting fraudulent transactions from highly imbalanced datasets.

Random Forests

Table 2 Random Forests Hyperparameter

Hyperparameter	Value	Description
n_estimators	100	Number of trees in the forest
min_samples_split	10	Minimum samples to split an internal node
min_samples_leaf	1	Minimum samples required to form a leaf node
max_leaf_nodes	None	No restriction on the number of leaf nodes
oob_score	TRUE	Enables Out-of-Bag validation scoring
n_jobs	-1	Use all available CPU cores for parallel processing
random_state	1	Ensures reproducibility of results

The `n_estimators` parameter, set to 100, determines the number of decision trees in the ensemble. This value provides a balance between accuracy and computational cost. Empirical testing showed that increasing the number of trees beyond 100 provided minimal performance improvement while significantly increasing computation time.

To reduce overfitting, `min_samples_split` was set to 10, ensuring that nodes are split only when there are sufficient samples to support the split. Similarly, `min_samples_leaf` was set to 1, which allows the model to capture all possible patterns, particularly given the limited number of fraudulent transactions in the dataset. Together, these parameters prevent the model from growing excessively complex trees while retaining its ability to learn from the available data.

The parameter `max_leaf_nodes` was left as `None` to allow unrestricted growth in the number of leaf nodes. This ensures that the decision trees have the flexibility to fully learn the data's patterns without being prematurely limited.

To provide an unbiased performance estimate, `oob_score` (Out-of-Bag score) was enabled. Out-of-Bag scoring uses the bootstrap sampling method inherent to Random Forest to validate the model on unseen data points, making it an effective method for evaluating generalization without requiring additional validation data.

The `n_jobs` parameter was set to -1, allowing the model to utilize all available CPU cores for parallel processing. This significantly reduces training time, especially when dealing with larger datasets or a higher number of decision trees. Finally, `random_state` was fixed at 1 to ensure reproducibility of results across different runs of the model.

In summary, the combination of these hyperparameters ensures that the Random Forest model achieves robust performance while avoiding overfitting and maintaining computational efficiency. The use of Out-of-Bag scoring provides further confidence in the model's generalization capability, making it well-suited for fraud detection tasks.

XGBoost

Table 3 XGBoost Hyperparameter

Hyperparameter	Value	Description
n_estimators	5000	Number of boosting rounds
max_depth	30	Maximum depth of each decision tree
learning_rate	0.01	Scaling factor for tree contributions
use_label_encoder	FALSE	Disables automatic label encoding
eval_metric	'logloss'	Evaluation metric for model performance

Key Hyperparameters

- **n_estimators=5000:** This defines the number of boosting iterations (i.e., the number of decision trees built sequentially). A high number of estimators ensures the model learns complex patterns but can increase training time.
- **max_depth=30:** The maximum depth of each tree, controlling the model's ability to capture intricate relationships in the data. A depth of 30 allows the model to handle high-dimensional and non-linear patterns effectively but may require careful regularization to prevent overfitting.
- **learning_rate=0.01:** This learning rate (also called shrinkage) scales the contribution of each tree, ensuring the model learns incrementally. A smaller learning rate like 0.01 helps achieve higher accuracy but requires a larger number of trees (n_estimators) for convergence.
- **use_label_encoder=False:** This disables the label encoding step, as the dataset may already have binary labels (0 for non-fraud, 1 for fraud), ensuring compatibility with preprocessing steps.
- **eval_metric='logloss':** The evaluation metric specifies how model performance is measured. Logarithmic Loss (logloss) is chosen because it penalizes incorrect classifications more strongly, making it ideal for binary classification tasks like fraud detection.

The XGBoost model, implemented with these hyperparameters, is tailored for the complexities of fraud detection, emphasizing accurate predictions for minority-class instances (fraudulent transactions). These settings optimize the trade-off between precision and recall, ensuring reliable fraud detection in imbalanced datasets.

Logistic Regression

Table 4 Logistic Regression

Hyperparameter	Value	Description
max_iter	1000	Maximum number of optimization steps
Regularization	L2 (Default)	Applies L2 penalty to reduce overfitting

Tuning Rationale

- `max_iter=1000`: This value defines the maximum number of optimization iterations for convergence. A higher value ensures that the model can handle larger datasets and more complex decision boundaries without halting prematurely.
- `Regularization=L2`: L2 regularization prevents overfitting by penalizing large coefficients, ensuring that the model remains stable and interpretable, even when features are noisy or redundant.

Logistic Regression is an essential component of this study, offering an interpretable baseline for fraud detection tasks. Although limited in handling non-linear patterns, its performance is significantly enhanced when paired with data preprocessing techniques that address class imbalance. With proper tuning, Logistic Regression provides a reliable starting point for detecting fraudulent transactions in imbalanced datasets.

6. Evaluation and Analysis

3.1 Evaluation Metrics

Evaluation metrics play a vital role in measuring the performance of machine learning models, especially in imbalanced classification tasks like credit card fraud detection. This study emphasizes four essential metrics: Precision, Recall, F1-Score, and ROC-AUC. These metrics collectively offer valuable insights into the model's effectiveness in identifying fraudulent transactions while reducing false positives and false negatives.

Precision

Precision indicates the percentage of correctly classified fraudulent transactions out of all transactions predicted as fraud. It measures the model's accuracy in identifying actual fraud cases while reducing false positives. A high precision score means fewer legitimate transactions are mistakenly flagged as fraudulent, which is crucial in real-world scenarios to avoid unnecessary investigations and customer inconvenience.

$$Precision = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Positives\ (FP)}$$

Recall

Recall measures the proportion of actual fraudulent transactions that the model successfully detects. It emphasizes the model's capacity to identify as many fraud cases as possible. In fraud detection, achieving a high recall is critical, as failing to detect fraudulent transactions can result in significant financial losses.

$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Negative\ (FN)}$$

F1-Score

The F1-Score merges Precision and Recall into a unified metric by computing their harmonic mean. It is particularly valuable when working with imbalanced datasets, as it balances the trade-off between detecting fraud (Recall) and minimizing false positives (Precision). A higher F1-Score signifies a well-rounded performance, reflecting the model's ability to manage both aspects effectively.

$$F1 - Score = 2x \frac{Precision \times Recall}{Precision + Recall}$$

ROC-AUC (Receiver Operating Characteristic - Area Under the Curve)

The ROC-AUC metric evaluates a model's capacity to differentiate between fraudulent and non-fraudulent transactions across various classification thresholds. It measures the area under the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (Recall) against the False Positive Rate. A higher ROC-AUC score, nearing 1.0, indicates superior performance in distinguishing between the two classes.

$$ROC - AUC = \int_0^1 TPRd(FPR)$$

Where:

- True Positive Rate (TPR): Recall
- False Positive Rate (FPR): The proportion of non- fraudulent transactions incorrectly classified as fraud

Given the imbalanced nature of fraud detection datasets, metrics like Precision, Recall, and F1-Score provide a more meaningful evaluation compared to overall accuracy, which can be misleading. For instance, a model predicting all transactions as non-fraudulent in a dataset with 99% non-fraud instances would achieve high accuracy but fail to detect fraud. By combining these metrics, we ensure the model's effectiveness in identifying fraud while minimizing false alarms.

3.2 Model Performance and Results from SMOTE (Over Sampling)

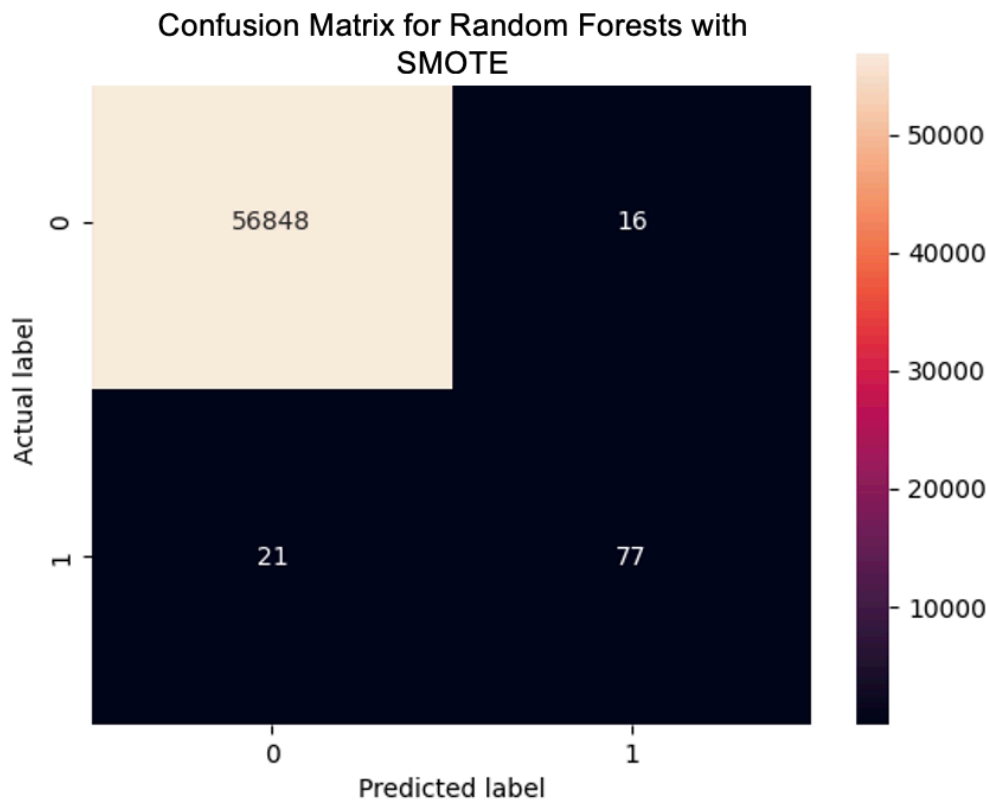


Figure 11 Confusion Matrix for Random Forest with SMOTE

Random Forest:

- True Positives (TP): 77 fraud cases correctly identified.
- False Positives (FP): 16 legitimate transactions flagged as fraud.
- True Negatives (TN): 56,848 legitimate transactions correctly classified.
- False Negatives (FN): 21 fraud cases missed.

Random Forest achieves the highest Precision among the models, minimizing false alarms with only 16 false positives while maintaining a Recall of 78.6%. This balance results in a strong F1-Score, reflecting its ability to detect fraud effectively while keeping false positives at a manageable level. The low number of false alarms ensures that customers' legitimate transactions are rarely flagged, making Random Forest well-suited for fraud prevention systems where minimizing customer disruptions is critical. Its ensemble learning approach, combining multiple decision trees, enhances its robustness and stability, contributing to its reliable performance in detecting fraud even under imbalanced data conditions.

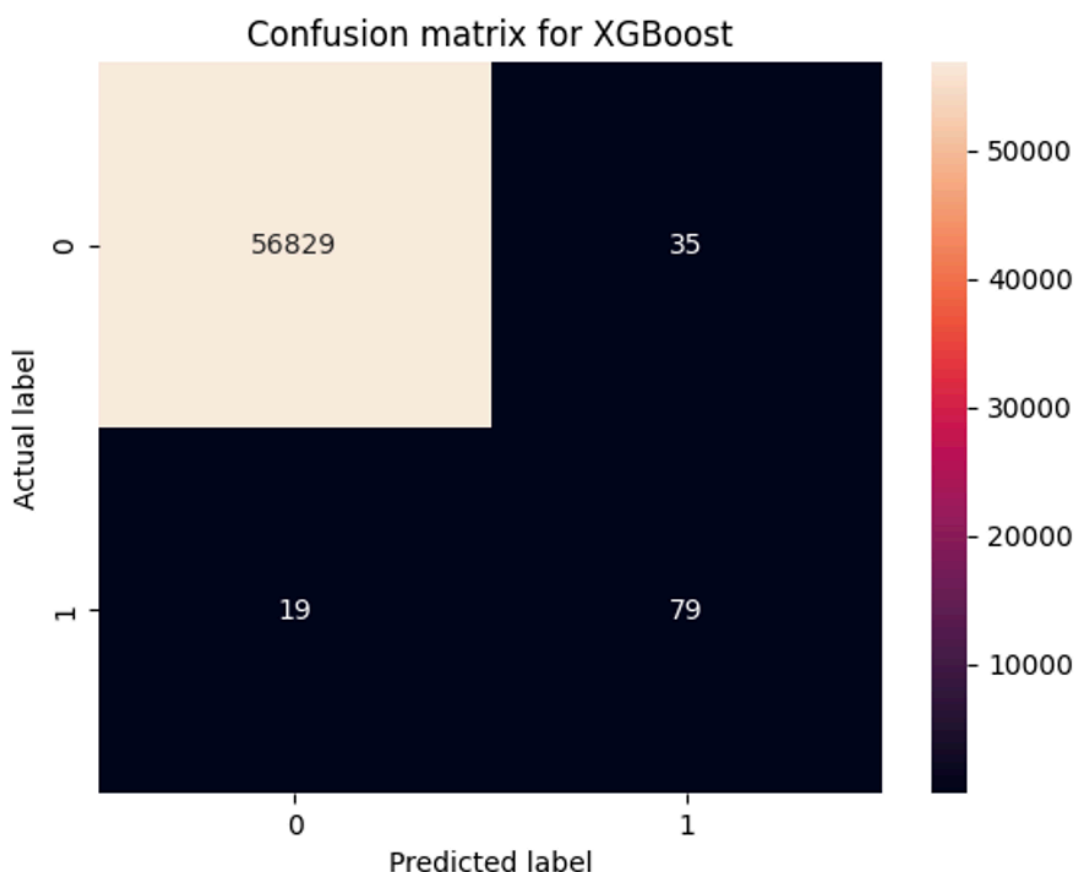


Figure 12 Confusion Matrix for XGBoost with SMOTE

XGBoost:

- True Positives (TP): 79 fraud cases correctly identified.
- False Positives (FP): 35 legitimate transactions misclassified as fraud.
- True Negatives (TN): 56,829 legitimate transactions correctly classified.
- False Negatives (FN): 19 fraud cases missed.

XGBoost demonstrates a well-rounded performance by correctly identifying 79 fraud cases (TP) while generating only 35 false positives (FP). Its Recall of 80.6% indicates its strong ability to detect fraudulent transactions, while its relatively low number of false positives supports a solid Precision. The resulting F1-Score confirms its capacity to balance fraud

detection and minimizing incorrect classifications. XGBoost's use of gradient boosting enables it to iteratively improve performance by focusing on difficult-to-classify instances, making it highly adaptive to fraud patterns in complex, real-world datasets. Despite being computationally intensive, its ability to detect even subtle fraud patterns makes it highly effective for large-scale fraud monitoring systems.

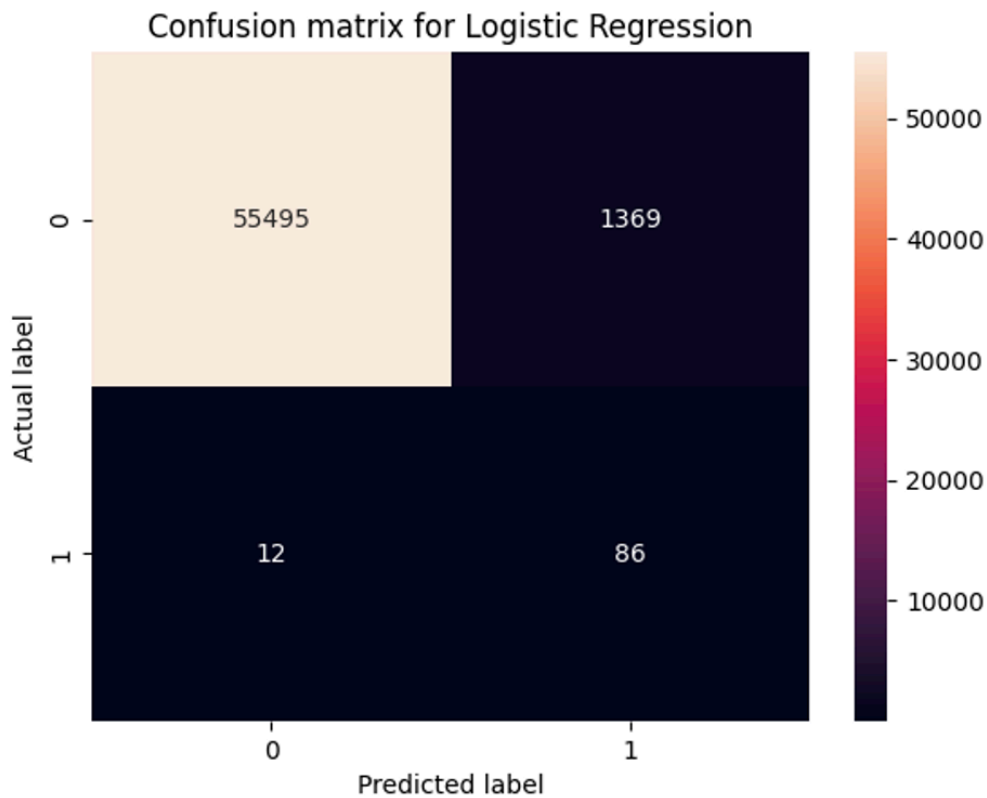


Figure 13 Confusion Matrix for Logistic Regression with SMOTE

Logistic Regression:

- True Positives (TP): 86 fraud cases correctly identified.
- False Positives (FP): 1,369 legitimate transactions incorrectly flagged as fraud.
- True Negatives (TN): 55,495 legitimate transactions correctly classified.
- False Negatives (FN): 12 fraud cases missed.

Logistic Regression achieves the highest Recall (88%), identifying 86 fraud cases (TP) while missing only 12 fraud cases (FN). However, its extremely low Precision is driven by a staggering 1,369 false positives (FP), making it less suitable for real-world applications where minimizing false alarms is essential. The high false positive rate could overwhelm fraud investigation teams, reducing operational efficiency. Despite its limitations, Logistic Regression's linear decision boundary makes it interpretable and useful as a baseline model for initial fraud detection assessments. Its strong Recall demonstrates its capability to capture

nearly all fraudulent transactions, highlighting its potential as part of a hybrid detection system when paired with more precise models.

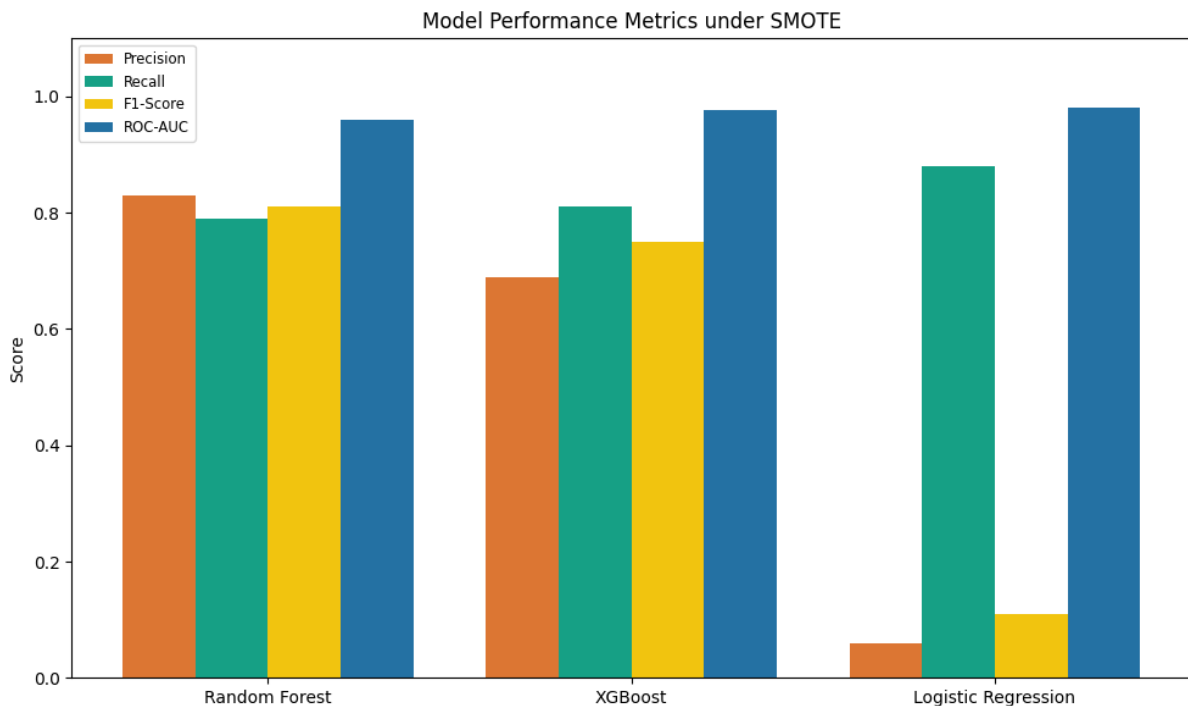


Figure 14 Model Performance Metrics under SMOTE

Under SMOTE (Synthetic Minority Oversampling Technique), Random Forest and XGBoost emerge as the most effective models for fraud detection by achieving a well-balanced trade-off between Precision, Recall, and overall performance. SMOTE enhances the representation of the minority class (fraudulent transactions), enabling these models to identify fraud more accurately without being overwhelmed by the class imbalance inherent in the dataset.

Random Forest stands out for its high Precision, minimizing false positives (16) while maintaining a respectable Recall of 78.6%. This makes it ideal for applications where avoiding false alarms is critical, such as real-time fraud prevention systems where false positives can disrupt customer transactions and damage user trust. Its ensemble-based structure enhances its stability and predictive accuracy by aggregating predictions from multiple decision trees, ensuring consistent performance even with imbalanced data.

XGBoost, known for its adaptive boosting mechanism, achieves a high Recall (80.6%) with relatively low false positives (35). This balance makes it suitable for fraud detection scenarios where missing fraudulent transactions poses a significant financial risk. Its ability to correct classification errors iteratively ensures that even hard-to-detect fraud cases are recognized. Despite its higher computational complexity, XGBoost's effectiveness justifies its use in large-scale fraud detection systems requiring continuous monitoring.

Logistic Regression, while achieving the highest Recall (88%), struggles with an extremely low Precision due to 1,369 false positives. This makes it less practical for real-world deployment, where minimizing false alarms is essential to avoid overwhelming fraud investigation teams. The model's linear nature limits its ability to handle complex fraud patterns, explaining its weaker performance compared to ensemble models like Random Forest and XGBoost.

Overall, applying SMOTE successfully addresses the class imbalance by improving the detection of fraudulent transactions. However, ensemble models like Random Forest and XGBoost outperform Logistic Regression due to their superior capacity to balance Precision and Recall, making them more suitable for deployment in fraud detection frameworks. In real-world applications, selecting a model depends on the specific business requirements, such as prioritizing customer experience or minimizing financial losses. Combining these models through hybrid approaches could further enhance fraud detection accuracy while maintaining operational efficiency.

3.3 Model Performance and Results from Tomek Links (Under Sampling)

Random Forest

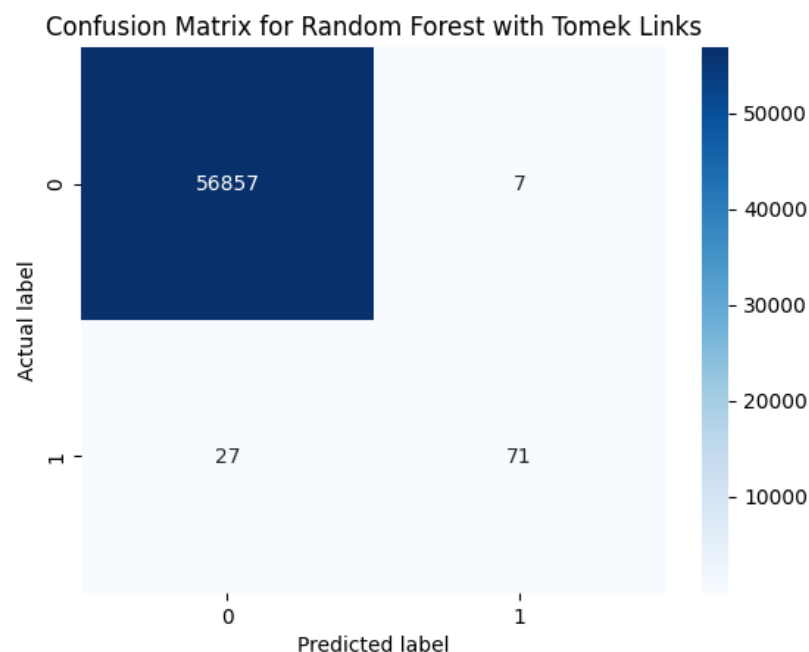


Figure 15 Confusion Matrix for Random Forest with Tomek Links

- True Positives (TP): 71
- False Positives (FP): 7
- True Negatives (TN): 56,857
- False Negatives (FN): 27

Random Forest demonstrates high Precision (91%), indicating its strong ability to correctly identify fraud while keeping false positives low (7 legitimate transactions misclassified).

Its Recall (72%) shows that it successfully detects the majority of fraudulent cases, though it misses 27 fraud cases (FN). This balance results in a strong F1-Score (0.81), reflecting its ability to handle fraud detection effectively while minimizing false alarms. Its ROC-AUC (0.94) underscores its robust capacity to distinguish between fraudulent and legitimate transactions. However, its slightly lower Recall compared to XGBoost suggests room for improvement in identifying difficult-to-detect fraud patterns.

XGBoost Using Tomek Links

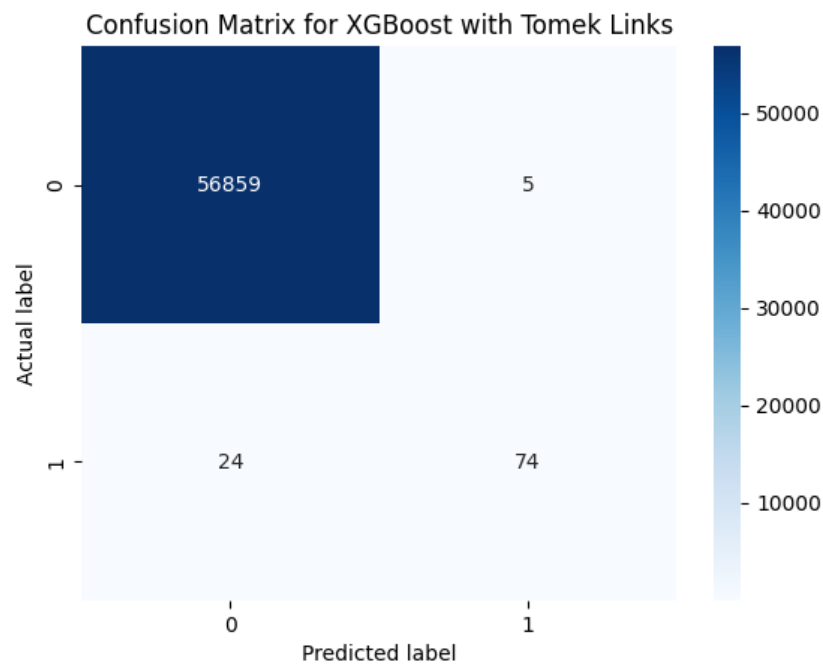


Figure 16 Confusion Matrix for XGBoost with Tomek Links

- True Positives (TP): 74
- False Positives (FP): 5
- True Negatives (TN): 56,859
- False Negatives (FN): 24

XGBoost achieves the highest Precision (94%), correctly identifying fraudulent transactions while minimizing false positives (5 cases). Its Recall (76%) surpasses that of Random Forest, indicating a better ability to detect fraud. Despite missing 24 fraud cases (FN), its high F1-Score (0.84) highlights its effectiveness in balancing fraud detection with minimizing false alarms. Its exceptional ROC-AUC (0.98) indicates near-perfect discrimination between fraudulent and non-fraudulent transactions. This combination of high Precision, Recall, and F1-Score makes XGBoost a top-performing model, suitable for real-world fraud detection scenarios.

Logistic Regression Using Tomek Links

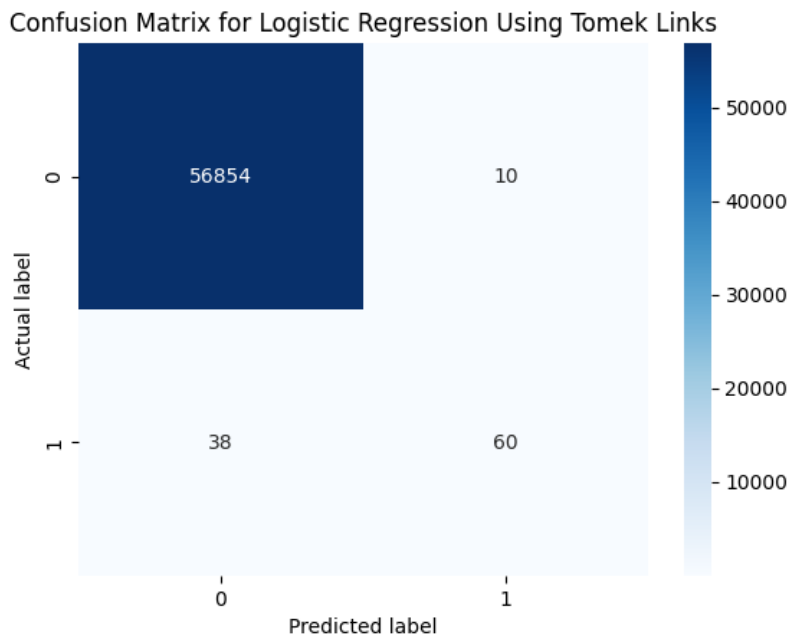


Figure 17 Confusion Matrix for Logistic Regression with Tomek Links

- True Positives (TP): 60
- False Positives (FP): 10
- True Negatives (TN): 56,854
- False Negatives (FN): 38

Logistic Regression demonstrates moderate performance under Tomek Links. Its Precision (86%) is competitive, showing its capacity to correctly identify fraud with a relatively low number of false positives (10 cases). However, its Recall (61%) indicates that it misses 38 fraud cases, a significant drawback in high-stakes fraud detection environments. Its F1-Score reflects this imbalance, being notably lower than that of XGBoost and Random Forest. Despite this, its high ROC-AUC (0.97) shows that it can effectively separate classes when threshold adjustments are applied. However, its inability to maintain a better Precision-Recall balance makes it less suitable for operational fraud detection tasks.

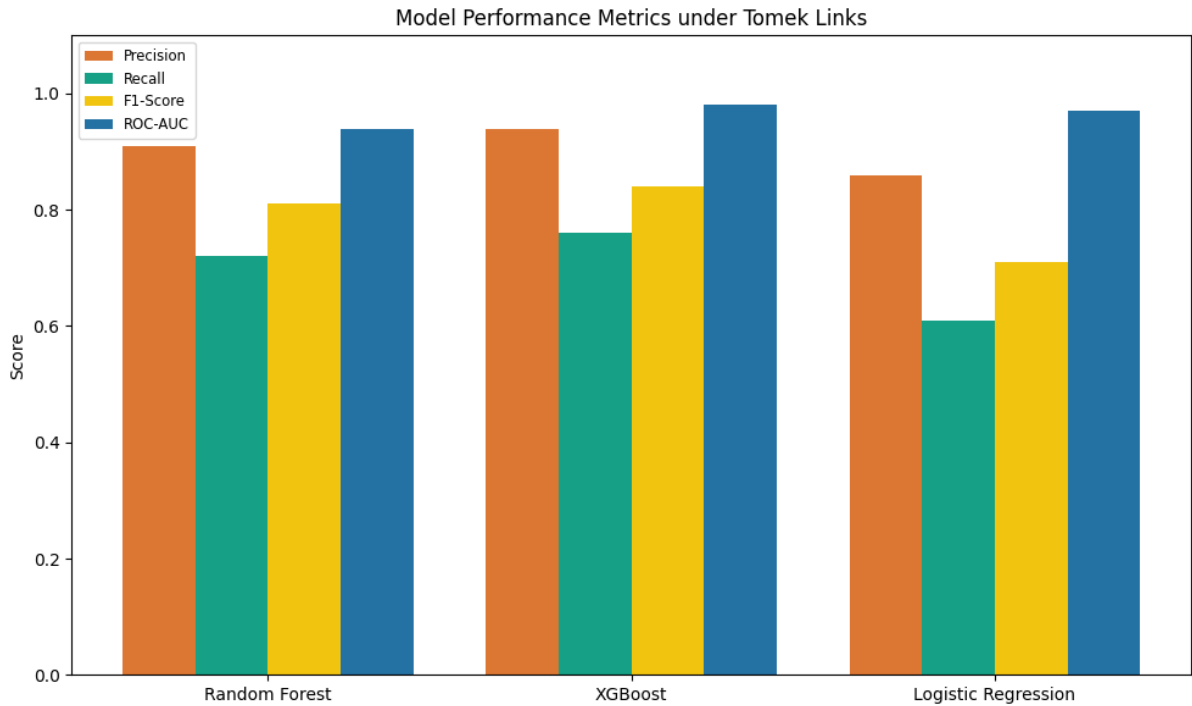


Figure 18 Model Performance Metrics under Tomek Links

Among the three models evaluated under Tomek Links, XGBoost demonstrates the strongest overall performance, achieving the highest Precision (94%) and Recall (76%). This indicates that XGBoost effectively minimizes false positives while detecting a significant proportion of fraudulent transactions, making it highly reliable for fraud detection tasks. Random Forest follows closely, with a Precision of 91% and Recall of 72%, showing strong predictive power and a balanced ability to classify legitimate and fraudulent transactions. In contrast, Logistic Regression achieves a Precision of 86% but lags in Recall at just 61%, missing a notable number of fraud cases. This limitation reduces its effectiveness in scenarios where high Recall is critical.

The F1-Score, which balances Precision and Recall, further underscores XGBoost's superior performance (0.84), with Random Forest achieving a slightly lower score of 0.81. Logistic Regression's F1-Score of 0.71 reflects its struggle to achieve a balance between detecting fraud and avoiding false positives. Additionally, all three models achieve strong ROC-AUC values, with XGBoost leading at 0.98, indicating excellent overall classification ability, followed by Logistic Regression (0.97) and Random Forest (0.94).

Conclusions

Overview

Detecting credit card fraud in highly imbalanced datasets remains a critical challenge for financial institutions due to the low proportion of fraudulent transactions compared to legitimate ones. This study explored how different machine learning models—Logistic Regression, Random Forest, and XGBoost—performed when paired with two resampling techniques: SMOTE (oversampling) and Tomek Links (undersampling). The goal was to evaluate how these models and sampling methods enhance fraud detection accuracy while minimizing false positives, ensuring balanced Precision and Recall for real-world applications.

Summary of the investigation study

The performance of the models under Tomek Links and SMOTE sampling techniques highlights the nuanced interplay between resampling methods and model capabilities in fraud detection. The results reveal why certain models outperformed others under specific conditions. XGBoost consistently emerged as the best-performing model, particularly under Tomek Links. This can be attributed to its ability to handle complex relationships in data through gradient boosting. XGBoost effectively adapts to imbalanced datasets by optimizing its learning process for the minority class, even with minimal undersampling as in Tomek Links. Its tree-based ensemble structure also allows it to capture intricate patterns in fraudulent transactions, leading to superior Precision and Recall compared to other models. SMOTE further enhanced its performance by balancing the dataset, enabling the model to identify more fraudulent cases while maintaining a low false-positive rate.

Random Forest also demonstrated strong performance, especially in terms of Precision. Its ability to minimize false positives under Tomek Links indicates that the model is highly effective at leveraging the cleaner majority-class data produced by this undersampling technique. However, compared to XGBoost, Random Forest is slightly less adept at capturing the subtle complexities in fraud detection, resulting in marginally lower Recall and F1-Scores. Under SMOTE, Random Forest benefited from the synthetic minority samples, although its reliance on bootstrapping might limit its ability to generalize as effectively as XGBoost.

Logistic Regression struggled to compete with the ensemble models due to its linear nature. While it performed well in terms of Recall under SMOTE, its high false-positive rate led to poor Precision, which significantly impacted its overall F1-Score. Under Tomek Links, the model failed to leverage the undersampled data effectively, reflecting its limited ability to learn non-linear patterns that are often critical in fraud detection tasks.

In summary, ensemble models like XGBoost and Random Forest excelled under both sampling techniques due to their capacity to handle class imbalance and capture complex fraud patterns. XGBoost stands out for its adaptability and consistent performance across Precision, Recall, and F1-Score. Logistic Regression, while computationally efficient, was less effective in handling the intricacies of fraud detection, especially in imbalanced datasets. This underscores

the importance of selecting models that align well with the properties of the resampling technique used.

Recommendations (Business-Oriented)

Operational Efficiency:

Random Forest with Tomek Links is recommended for businesses seeking to minimize false positives while maintaining customer trust. This configuration ensures high Precision, reducing legitimate transactions being flagged as fraud. It is ideal for real-time payment systems, online marketplaces, and subscription management platforms, where frequent false alerts could drive away customers or disrupt recurring payments.

Maximizing Fraud Detection:

XGBoost with SMOTE is best for high-risk financial services, where maximizing Recall is critical to identifying the maximum number of fraudulent cases. This configuration is suitable for credit monitoring services, anti-fraud compliance checks, and transaction surveillance systems used in financial institutions. Businesses dealing with large-scale transactions should consider using distributed computing infrastructure to ensure seamless real-time fraud detection at scale.

Compliance and Transparency:

Logistic Regression should be used as a baseline model or in compliance-driven environments where interpretability and transparency are essential. This makes it suitable for regulatory audits, internal fraud investigations, and environments requiring explainable models to meet GDPR or data privacy regulations. Businesses can enhance its performance by integrating feature engineering techniques such as interaction terms or polynomial expansions to improve its predictive capabilities.

Areas for Future Business-Focused Research

Hybrid Sampling Techniques:

Investigate combining SMOTE with Tomek Links to balance datasets while reducing noise, ensuring higher Precision and Recall for fraud detection systems. This could enable businesses to build adaptive fraud detection frameworks that adjust dynamically based on current fraud trends.

Advanced Model Architectures:

Explore Neural Networks, Autoencoders, and LSTMs for time-series fraud detection. Businesses involved in e-commerce or financial trading could benefit from models that analyze historical transaction sequences for unusual activity patterns.

Real-Time Fraud Prevention Systems:

Evaluate the models' performance in a real-time detection environment by deploying them through cloud-based services like AWS SageMaker, Google AI Platform, or Azure ML. This would allow companies to detect fraud with minimal latency while scaling services globally.

Cost-Sensitive Learning:

Consider models that factor in business costs associated with false positives (customer complaints) and false negatives (financial loss). This could help businesses implement more strategic fraud management systems by calculating the net business impact of fraud detection errors.

Personal Evaluation

This project provided valuable insights into the challenges and solutions for fraud detection in imbalanced datasets, emphasizing the need for carefully selecting models, evaluation metrics, and resampling techniques to improve detection accuracy. The study reinforced the importance of applying appropriate machine learning algorithms that align with specific business and operational goals, such as minimizing false positives or maximizing fraud detection rates. Through detailed model comparisons, I gained a deeper understanding of how ensemble methods like Random Forest, and XGBoost outperform traditional linear models such as Logistic Regression, particularly in complex fraud detection scenarios.

The research process enhanced my knowledge of key machine learning concepts, including model training, hyperparameter tuning, and evaluation metrics like Precision, Recall, F1-Score, and ROC-AUC. I learned how models respond to different resampling techniques, such as SMOTE and Tomek Links, and how these methods impact fraud detection by balancing datasets and improving class separation. This experience taught me how to build, evaluate, and fine-tune models in a structured and analytical manner.

Additionally, the project strengthened my data preprocessing skills, including feature engineering, dataset splitting, removing outliers, and handling. I realized how crucial these steps are for achieving accurate predictions, especially when dealing with sensitive financial data. Performing correlation analysis and selecting relevant features also deepened my understanding of feature importance in predictive modelling.

From a business perspective, the project highlighted the trade-offs between different evaluation metrics. For example, while Recall is critical for detecting fraud, a high false-positive rate can disrupt business operations and negatively impact customer experience. This realization emphasized the importance of balancing business costs and operational efficiency when recommending machine learning models for real-world deployment.

Moreover, interpreting model results and making strategic recommendations for business operations required critical thinking and a strong analytical mindset. I learned how to analyze evaluation metrics, adjust model thresholds, and consider how models perform under various operational constraints. This experience reinforced the need for continuous model monitoring and retraining to adapt to evolving fraud patterns.

Overall, this project significantly expanded my technical expertise in machine learning, fraud detection, and data science workflows. It also improved my ability to translate technical findings into actionable business insights, a skill crucial for real-world data-driven decision-making. This experience prepared me for applying advanced machine learning concepts in

professional environments while balancing technical performance metrics with business objectives.

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